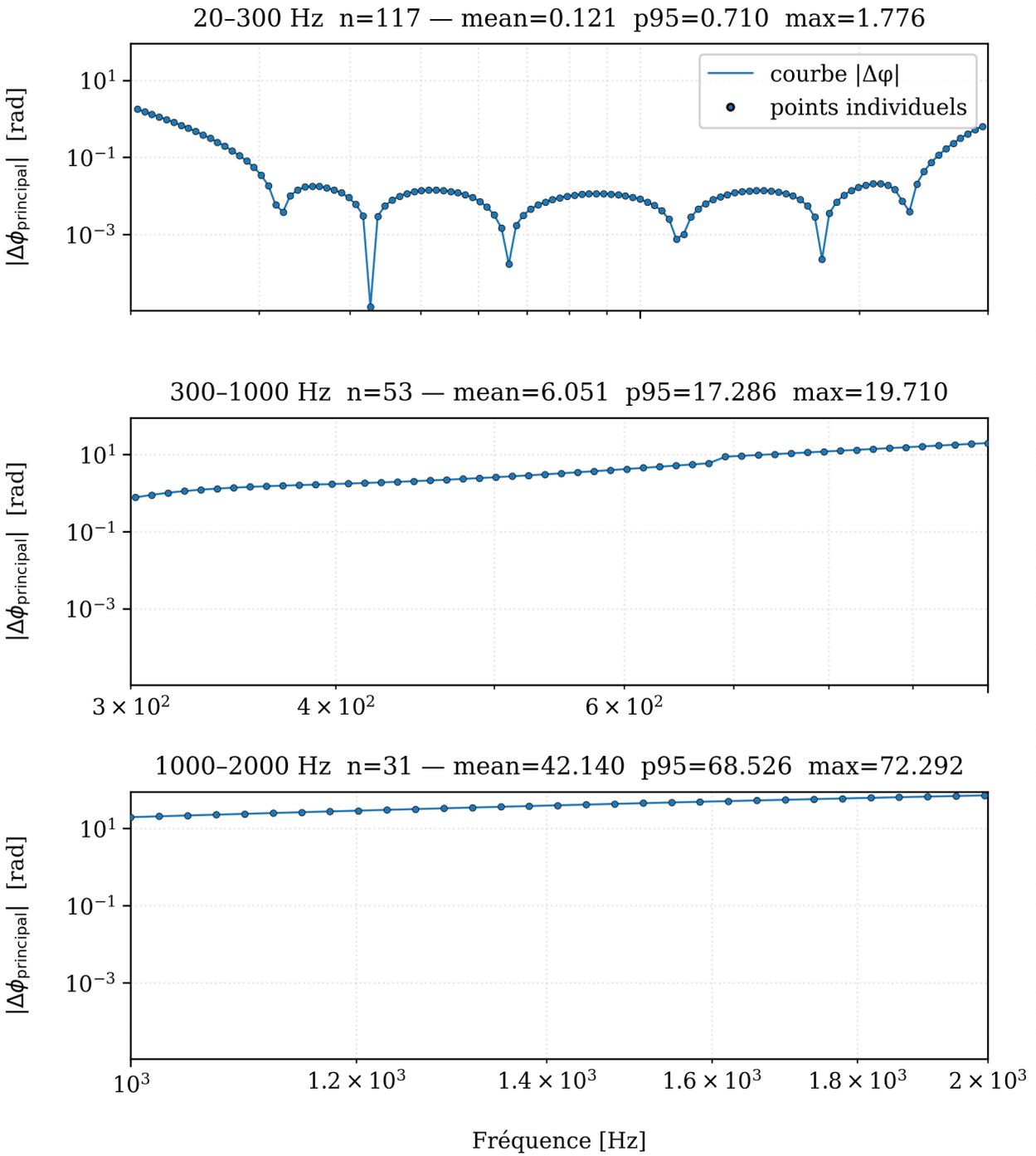
Phase Residual $|\Delta\phi|$ by Frequency Band (ϕ_{ref} vs ϕ_{MCGT})



Statistiques globales (20–300 Hz)
mean = 0.121 rad
p95 = 0.710 rad
max = 1.776 rad

Bandes individuelles
20–300 Hz : n=117, mean=0.121, p95=0.710, max=1.776
300–1000 Hz : n=53, mean=6.051, p95=17.286, max=19.710
1000–2000 Hz : n=31, mean=42.140, p95=68.526, max=72.292

Méta (résumé)
ini: zz-configuration/gw_phase.ini
params: {'m1': 30.0, 'm2': 25.0, 'q0star': 0.0, 'alpha': 0.0, 'phi0': 0.0, 'tc': 0.0, 'tol': 1e-06}
reference: {'csv': 'zz-data/chapter09/09_phases_imrphenom.csv', 'meta': 'zz-data/chapter09/09_phases_imrphenom.meta.json', 'source_tag': 'file_existing'}
grid_used: {'fmin_Hz': 10.0, 'fmax_Hz': 2041.7379446695293, 'dlog10': 0.01, 'n_points_used': 232}
metrics_window_Hz: [20.0, 300.0]
metrics_active: {'mean_abs_20_300': 26.831599861688545, 'max_abs_20_300': 93.7503296126555, 'p95_abs_20_300': 81.29194862161798, 'variant': 'calibrated'}
metrics_raw: {'mean_abs_20_300': 29.13754874877777, 'max_abs_20_300': 141.7454716684315, 'p95_abs_20_300': 110.38826055552936}
metrics_cal: {'mean_abs_20_300': 26.831599861688545, 'max_abs_20_300': 93.7503296126555, 'p95_abs_20_300': 81.29194862161798}
calibration: {'enabled': True, 'mode': 'phi0,tc', 'model_used': 'phi0_tc', 'phi0_hat_rad': -58.94512065649389, 'tc_hat_s': 0.07323215807767205, 'initial_window_Hz': [20.0, 300.0], 'used_window_Hz': [30.0, 250.0], 'weight': '1/f2', 'auto_tightened': True, 'tighten_threshold_p95_rad': 5.0, 'p95_check_before_rad': 164.41347534829, 'p95_check_after_rad': 81.29194862161798, 'n_points': 92}
outputs: {'phases_mcgt_csv': 'zz-data/chapter09/09_phases_mcgt.csv', 'diff_phase_csv': None, 'comparison_milestones_csv': None, 'fisher_scan2D_csv': None}
repro: {'git_hash': '9c243cef287d57af98025112b4fe96b6740d7909', 'python': '3.12.11', 'libs': {'numpy': '2.3.5', 'pandas': '2.3.3'}}